**25901184 – Scheme of Evaluation**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**
**MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026**
**Fourth SEMESTER**
**Skill Enhancement Course (SEC) - MG4SECBCA200 - OBJECT ORIENTED PROGRAMMING USING JAVA**
**(2024 ADMISSION ONWARDS)**

**Duration: 1 hour 30 minutes** **Maximum Marks: 50**

## Part A Very Short Answer Questions. Answer all questions – 3 Marks each

1. What is Java Virtual Machine (JVM)? [K] / [CO1]

JVM is a virtual machine that enables the execution of Java bytecode. it acts as an interpreter between the Java programming language and the underlying hardware. It provides a runtime environment for Java applications to run on different platforms and operating systems. 3 marks.

2. How does a switch statement simplify multiple decision-making in a program? [U] /[CO1]

Defines a switch statement as a control structure used to select one block of code from multiple alternatives -1 Mark, Explanations that it replaces multiple if-else-if conditions, making the code more organized and easier to read- 2 marks

3. How will you check whether a Vector is empty or not? [An] / [CO2]

Using the isEmpty() method or by checking its size using vector.size(). Explanation of any one method with Java statement– 3 marks

4. Explain abstract methods in Java [An] / [CO2]

A method declared without an implementation (or a body) : Definition - 1 Mark, Example in Java -2 Marks

5.Define a package [U] / [CO3] - a mechanism used to group related classes, interfaces, enumerations, and sub-packages into a single namespace or unit: Definition - 3 marks

**[3×5 = 15]**

## Part B Short Answer Questions. Answer any 4 questions- 5 Marks each

6 Compare static variables and instance variables. [U] / [CO1]

-static variables (class-level variables shared by all objects)., instance variables (object-level variables.unique to each object). Definition- 2 marks. (static variables stored once per class; instance variables created separately for each object, static variables accessed using class name; instance variables accessed through objects) Highlight key differences with example - 3 marks

7. Describe method overloading in Java and demonstrate its implementation with a suitable example. [U] / [CO1]

Definition – 2 Marks, implementation with suitable example in java -3 marks

8. Create and display elements of an n × n matrix [An] / [CO2]

Program with correct syntax (or explanation using proper Java Statements with correct syntax- 5 Marks,

9.Explain how to prevent overriding a variable in Java. Give an example. [U] / [CO2]

-using the final keyword – explanation 2 marks, Example - 3 Marks

10. Explain the steps involved in creating a thread by extending the Thread class with a simple example. [An] /[CO3]

Explanation of the steps - 3 marks, Example - 2 Marks

**[5×4 = 20]**

## Part C Essay Questions Answer any 1 question

11. With the help of suitable examples explain Wrapper methods. [A] / [CO2]

Explain any three wrapper methods such as parseXxx(), valueOf(), toString(), and xxxValue() with proper explanation and example in Java – 15 Marks. (or Definition and purposes of wrapper classes in Java, explanation and examples of commonly used wrapper classes and short description of wrapper methods – give a maximum of 10 marks).

12. Create a class Student with attributes roll no, name, age and course. If age of student is not in between 15 and 21 then generate user-defined exception "AgeNotWithinRangeException". [A] / [CO3]

Class definition with proper attributes and methods, constructors to methods to initialize and display student details - 8 marks, Define AgeNotWithinRangeException (extending Exception), Implement age validation logic (checks whether age is between 15 and 21), throw the exception using throw keyword when the condition fails., Handle the exception using try-catch block- 7 Marks

**[15×1 = 15]**